

Vericoding

Verification-Driven Development

From behavioural features to machine-checked proofs

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Vibecoding

The status quo of LLM-driven development.

You prompt. The model emits. It runs.

What you leave behind:

- No specification of intended behaviour
- No evidence of correctness
- No conformance regime for the next generation

The only durable artifact is the code itself.

Vericoding

The methodology specsaver operationalises.

You write a specification. The spec survives.

What you leave behind:

- Features — Gherkin scenarios domain experts can review
- Contracts — mathematical pre/post-conditions
- Proof obligations — machine-checked in Rocq/Coq

Generated code is a replaceable implementation detail.

Two Development Flows

Retrospective

- Feature — Gherkin scenario tables
- Implementation — write the production code
- Testing (sanity) — make sure it basically works
- Contract — capture observed behaviour as mathematical spec
- Testing (dialectic) — re-run under contract checking; either contract or implementation is wrong — the dialectic
- Formal Proof — lower to Coq, LLM closes obligations; DISPROVED → back to dialectic

Contract-First

- Feature — Gherkin scenario tables
- Contract — write spec as acceptance criteria
- Implementation — build code against the specification
- Testing — validate that implementation satisfies contract
- Formal Proof — machine-checked against kernel

Both converge: the contract is the lasting artifact.
DISPROVED yields witnesses → back to the dialectic.

Contract Language

The pure terminating fragment
of Python

Every clause is a boolean-valued lambda:

- `requires(state, args)` — admissibility
- `ensures(state, args, result, new_state)` — success post
- `when(state, args)` — exception conditions
- `invariant(state)` — global legality

Static purity check rejects side effects, loops, and non-boolean returns before execution.

Anatomised as mathematical
record

$$C = \langle \Sigma, \text{args}, \text{pre}, \text{post}, \\ \mathcal{X}, \mathcal{I}, \mathcal{D}, \mathcal{W}, \mathcal{R}, \mathcal{G} \rangle$$

$\mathcal{X}$ — exceptional exits (when + ensures + frame)

$\mathcal{I}$ — invariants on every state

$\mathcal{D}$ — derived-state definitions

$\mathcal{W} / \mathcal{R}$ — semantic frame

$\mathcal{G}$ — ghost state

Contract Anatomy: Implementation

A simple inventory reserve function against SQLAlchemy:

```
def reserve(self, engine, sku, order_id, quantity):
    with engine.begin() as conn:
        row = conn.execute(
            text("SELECT on_hand, reserved FROM products"
                " WHERE sku = :sku"), {"sku": sku}
        ).fetchone()

        if row is None: raise ProductNotFoundError(sku)

        on_hand, reserved = row
        available = on_hand - reserved

        if available < quantity:
            raise InsufficientStockError(sku, quantity, available)

        conn.execute(
            text("UPDATE products SET reserved = reserved + :qty"
                " WHERE sku = :sku"),
            {"qty": quantity, "sku": sku},
        )

    return ReservationReceipt(sku=sku, quantity=quantity)
```

Contract Anatomy: Specification

The same operation, specified once:

```
reserve_contract = Contract(  
    requires=[  
        lambda s, a: a.quantity > 0,  
        lambda s, a: a.sku in s.observed.products,  
    ],  
    ensures=[  
        lambda s, a, r, s2: (  
            s2.products[a.sku].reserved == s.products[a.sku].reserved + a.quantity),  
    ],  
    exceptions=[  
        ExcExit(raises=InsufficientStockError,  
            when=[lambda s, a:  
                s.products[a.sku].on_hand - s.products[a.sku].reserved < a.quantity],  
            writes={"state.failure_log"}),  
    ],  
    writes={"state.products[sku].reserved", "state.failure_log"},  
    reads={"state.products[sku].on_hand", "state.products[sku].reserved"},  
)
```

requires
admissibility

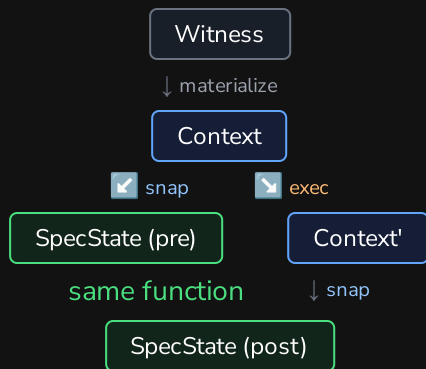
ensures
what changes

exceptions
when + outcome

writes/reads
semantic frame

Symmetric Projection

The commuting diagram. One projection `snap`, applied before and after execution. Contracts compare the pair.



Same predicates, two levels.

At runtime: the predicates evaluate as Python bool on real state — every Gherkin row is a concrete test.

At proof time: the same predicates are translated to Coq propositions over an abstract heap model — every obligation proved is a universal guarantee.

Symmetry as Bridge

Why do testing and proving agree? Because the state they see is produced by the same function.

Declared schema

The contract's ``state_schema`` names every observed field, its type, and its provenance. The runtime snapshot and the Coq state model are both derived from this single declaration.

Semantic frame

``writes`` and ``reads`` are declarative paths over the schema. The runtime checker and the frame-soundness proof enforce the same footprint.

Same snapshot

One function, called twice. No divergent "read" and "check" path. The frame, derived-consistency, and invariant checkers all operate over this shared interpretation.

The specification is the bridge. Write it once; test it on data today; prove it against the kernel tomorrow. The lift is automatic.

Symmetric Projection — Runner

1. materialize

witness → temp SQLite DB + engine + EventLog

2. pre-check

invariants on pre-state; admissibility (requires)

3. execute

service runs against real SQLite via SQLAlchemy;
wrapper emits typed events into the log

4. frame check

everything outside writes unchanged

5. derived check

derived fields $\equiv$ recomputed from observed

6. post-check

ensures (or exit ensures) for the outcome;
invariants on post-state

Domain Declaration

One object → runners, CLI, conformance suite.

```
inventory = SqlDomain(  
    name          = "inventory",  
    package       = "examples.inventory",  
    materializer  = SqlMaterializer(ddl, TableSpec[]),  
    projection    = SqlProjection(types, hooks),  
    operations    = [  
        SqlOperation(contract, wrapper,  
                        witness_builder, feature_file),  
        ...  
    ]  
)
```

scenario runners

CLI dispatch

conformance tests

Adding a domain: types + service + contracts + features + witness builders + one declaration. Everything else (materializer, projection, runner wiring, test dispatch) is derived.

Theory Stack

Services run unmodified on differentially-tested stubs.

Service ordinary SQLAlchemy / logging / OTel API code

Shim make_engine, make_log_capture, make_otel_capture

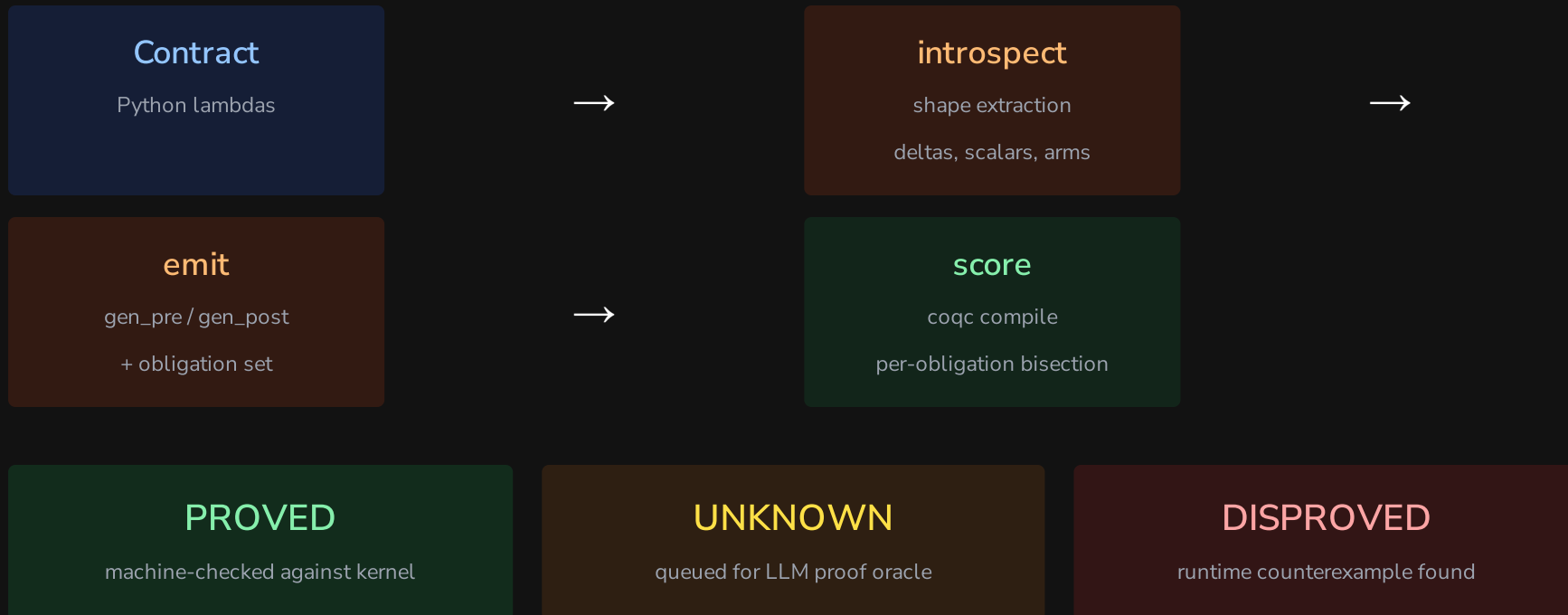
StubHandler operational semantics over the state model

State model TableStore / LogStore / OtelStore

Trace final interpretation (events)

Theory conformance: Stub and real library agree observationally on a differential suite; translator rejects out-of-fragment input loudly; state model and trace are pure data.

Lowering Pipeline



Example: Inventory Reserve

Pre-condition:

$$\text{pre}(s, a) = a.\text{quantity} > 0 \wedge a.\text{sku} \in s.\text{products}$$

Post-condition (excerpt):

$$\begin{aligned} \text{post}(s, a, r, s') &= s'.\text{products}[a.\text{sku}].\text{reserved} = s[\cdot].\text{reserved} + a.\text{quantity} \\ &\wedge \text{extends_by_one}(s.\text{reservation_log}, s'.\text{reservation_log}, \lambda e. e.\text{sku} = a.\text{sku}) \\ &\wedge \text{implies}(\text{crossed}(s, a, s'), \text{extends_by_one}(s.\text{alert_log}, s'.\text{alert_log}, \dots)) \end{aligned}$$

Exception exit:

$$\mathcal{X} = \{ \langle \text{InsufficientStock}, s.\text{available}(a.\text{sku}) < a.\text{quantity}, \{ \text{failure_log} \}, \text{failure_ensures} \rangle \}$$

Example: Invitations

Invite (insert frame):

$$\begin{aligned} \text{post}_{\text{invite}}(s, a, r, s') &= s'.\text{invitations}[a.\text{token}].\text{status} = \text{pending} \\ &\quad \wedge s'.\text{invitations}[a.\text{token}].\text{expires_at} = a.\text{now} + 7 \cdot 86400 \end{aligned}$$

Accept (multi-table delta):

$$\begin{aligned} \text{post}_{\text{accept}}(s, a, r, s') &= s'.\text{invitations}[a.\text{token}].\text{status} = \text{accepted} \\ &\quad \wedge s'.\text{members}[a.\text{user_id}].\text{org_id} = s.\text{invitations}[a.\text{token}].\text{org_id} \\ &\quad \wedge s'.\text{users}[a.\text{user_id}].\text{default_org} = s.\text{invitations}[a.\text{token}].\text{org_id} \end{aligned}$$

Design decisions for verifiability: Token is a caller-supplied argument (args-resolved insert key). **now** is an integer epoch argument (expiry is a pure comparison).

